UPEKSHA C. DISSANAYAKE

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EDUCATION

Doctor of Philosophy, Computational Chemistry Exp. Grad.: March 2026
Advisor: Dr. G. Andrés Cisneros **GPA 3.84**
The University of Texas at Dallas, Richardson, TX
Awards:

- Finalist of NVIDIA GPU Award, American Chemical Society, Fall Meeting, 2024
- Natural Science and Mathematics Conference Travel Award, The University of Texas at Dallas, June 2024
- Awardee of Graduate Research and Cancer Education (GRACE) Fellowship Program, University of Texas at Dallas, 2023

Master of Science, Industrial and Environmental Chemistry December 2019
University of Kelaniya, Sri Lanka **GPA 4.0**
Thesis Project: Investigation of the heavy metal contaminations and soil phosphate levels in agricultural lands in CKDu prevalence areas in North Central Province, Sri Lanka

Bachelor of Science, Major: Chemistry & Biochemistry, Minor: Botany February 2014 - March 2017
University of Kelaniya, Sri Lanka **GPA 3.40**
Honors/Awards/Scholarships: Dean's award for the academic excellence (2014-2015)

PROFESSIONAL EXPERIENCE

Computational Chemistry/ Bioinformatics Intern May 2025 – Present
Rayca Precision, 28 Geary St, San Francisco, CA

- Hands-on experience with AI-based molecular generation tools to design novel small molecules targeting disease-related proteins.
- Develop skills in managing large-scale molecular datasets and executing structure-based drug discovery workflows, including protein structure preparation, docking, molecular dynamics simulations, and property predictions.
- Gain experience integrating computational chemistry, bioinformatics, and AI models within an industry setting to address real-world challenges in therapeutics development

Graduate Research Assistant, The University of Texas at Dallas, Richardson, TX January 2022 – Present
Graduate Teaching Assistant
University of North Texas, Denton, TX January 2021 – December 2022
The Open University, Sri Lanka August 2019 – November 2020
University of Kelaniya, Sri Lanka July 2017 – December 2018

TECHNICAL SKILLS

- **Certifications**: Python Scripting for Molecular Docking; International workshop on Genome Informatics; Introduction to Big Data and Data Analytics

- **Protein Design & Generative Models:** Machine learning-based predictive models such as AlphaFold, Rosetta, ESMFold, ProteinMPNN, ColabFold, and OmegaFold.

- **Molecular Dynamics (MD):** GROMACS, AMBER, NAMD, CHARMM, OpenMM, TINKER; enhanced sampling, free energy calculations (FEP, TI, MBAR), MM/GB(PB)SA.

- **Quantum Mechanics (QM) & QM/MM:** Gaussian, Psi4, LICHEM; density functional theory (DFT), wavefunction/electronic analysis, polarizable/non-polarizable force field parameterization.

- **Docking & AI-Driven Drug Discovery:**
  AutoDock Vina, AutoDock 4, Schrödinger Glide, MOE, and AI-based frameworks (PocketFlow, MolMIM, DiffDock, and Boltz-2); ADMET prediction tools (ADMET-AI, pkCSM, SwissADME); structure- and ligand-based docking, virtual screening, pose prediction, binding affinity estimation, and integration with MD simulations.

- **Cloud Computing & HPC:** Microsoft Azure; containerized workflows (Docker, Singularity); HPC job schedulers (SLURM), large-scale automated pipelines for MD.

- **Programming & Data Science:** Python, PyTorch, R, Pandas, NumPy, SciPy, Scikit-learn, Matplotlib; RDKit, UNIX/Linux shell scripting, workflow automation.

- **Structural Analysis & Visualization:** VMD, PyMOL, UCSF Chimera, COOT, MDAnalysis, MDTraj, Open Babel, ChemDraw, GaussView, Avogadro, HyperChem, AIM, MultiWFN

- **Cancer Genome-Wide Association Study (GWAS):** PLINK; logistic regression, haplotype analysis, LD mapping, and SNP association modeling

RESEARCH EXPERIENCE

- Discovery and characterization of cancer biomarkers in NIH cancer databases (lung cancer, breast cancer, melanoma, prostate cancer, lymphoma, etc), focusing on DNA transaction genes, with applications of molecular dynamics (MD) and hybrid ab initio Quantum Mechanics/Molecular Mechanics (QM/MM) methods to enzyme studies.
- Investigation of the reaction mechanism of short-chain thioester hydrolysis by acyl hydrolase domains in trans-acyltransferase polyketide synthase enzyme using MD and QM/MM calculations (Collaboration with Dr. Matthew Jenner, University of Warwick, UK)
- Study of the effects of two cancer mutations associated with DNA glycosylase enzyme (MUTYH) on allosteric network interactions between DNA and the co-factor (Iron-Sulfur cluster) using MD simulations and analyses (Collaboration with Dr. Sheila David, University of California, Davis)
- Computational investigation of the structural features, functions, and reaction mechanism of Primase Polymerase C and CRISPR-associated Primase Polymerase enzymes, which are responsible for initiation of DNA replication and DNA repair. (Collaboration with Dr. Aidan Doherty, University of Sussex, UK) (Manuscript in preparation)
- Study of the effects of steric gate residues in 8-oxo guanosine incorporation by dCMP and dAMP in mitochondrial DNA polymerases. (Collaboration with Dr. Luis G. Brieba de Castro, Center for Research and Advanced Studies, Mexico)
- Application of diverse MD simulations and high-throughput methods and train a machine learning model to investigate the co-evolution dynamics of proteins within a bacterial toxin-antitoxin (ParD3-ParE3) system by studying non-interfacial, non-specific mutations. (Collaboration with Dr. Faruck Morcos, University of Texas at Dallas)

PUBLICATIONS

1. Molecular basis for short-chain thioester hydrolysis by acyl hydrolase domains in trans-acyltransferase polyketide synthases. *JACS Au, 2024,* https://doi.org/10.1021/jacsau.4c00837
2. Structure of human MUTYH and functional profiling of cancer-associated variants reveal an allosteric network between its [4Fe-4S] cluster cofactor and active site required for DNA repair. *Nature Communications, 2025,* https://doi.org/10.1038/s41467-025-58361-w
3. Computational Studies on the Functional and Structural Impact of Pathogenic Mutations in Enzymes. *Protein Science*, *2025,* https://doi.org/10.1002/pro.70081
4. A steric gate prevents mutagenic dATP incorporation opposite 8-oxo guanosine in mitochondrial DNA polymerases. *Federation of European Biochemical Societies Journal. 2025,* https://doi.org/10.1111/febs.70064

SELECTED PRESENTATIONS

Biophysical Society Annual Meeting	February 19 2025, February 11 2024
American Chemical Society Fall Meeting	August 20 2024, August 15 2023

PEER-REVIEW EXPERIENCE ORCID

Review activity for Public Library of Science One (*PLOS ONE*)